Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2024

Course : B.Tech
Code : 104705

Subject: Optical Fiber Communication

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct answer of the following (any seven question only): [2 x 7 = 14]

(a) Standard single mode fibers (SSMF) are utilized mainly for operation in _____
(i) C-band (ii) L-band
(iii) O-band (iv) C-band and L-band

(b) In single mode fibers, which is the most beneficial index profile?
(i) Step index (ii) Graded index
(iii) Coaxial cable (iv) Step and graded index

(c) _____ has more sophisticated structure than p-n photodiode.
(i) Avalanche photodiode
(ii) p-n junction diode
(iii) Zener diode
(iv) Varactor diode

(d) The optical source used for detection of optical signal is
(i) IR sensors (ii) Transistors
(iii) Zener diodes (iv) Photodiodes

(e) Polarization modal noise can _____ the performance of communication system.
(i) Degrade (ii) Improve
(iii) Reduce (iv) Attenuate

(f) Multimode step index fiber has a large core diameter of range is
(i) 100 to 300 µm (ii) 100 to 300 nm
(iii) 200 to 500 µm (iv) 200 to 500 nm

(g) What is a multi-functional element of optical network?
(i) Optical attenuation (ii) Wavelength
(iii) Optical node (iv) Hop

(h) In SONET, each synchronous transfer signal STS-n is composed of
(i) 2000 frames (ii) 4000 frames
(iii) 8000 frames (iv) 16000 frames

(i) The ring and star topologies are combined in a _____ configuration
(i) Mesh (ii) Fringe
(iii) Data (iv) Singular

(j) What is SDH?
(i) SDH is similar standard to SONET developed by ITU-T
(ii) Synchronous digital hierarchy
(iii) Synchronous digital hierarchy and is a similar standard to SONET developed by ITU-T
(iv) None of the mentioned

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Q.2

(a) Briefly explain the Attenuation in Optical Fiber Communication? [7]

(b) Differentiate between Single mode fibers and Multimode Fibers. [7]

Q.3

(a) Discuss the Mode theory for circular waveguides in Optical fiber. [7]

(b) Explain the Polarization? And Write done the different types of Polarization? [7]

Q.4

(a) Explain the block diagram of Optical Fiber Communication. And what are the typical applications of Optical Communication System? [7]

(b) Define the Optical fiber modes and its Con- figuration? [7]

Q.5

(a) Explain the Semiconductor Photodiode Detectors? [7]

(b) How does an Avalanche Photodiode Detectors differ from a Semiconductor Photodiode Detectors? [7]

Q.6

(a) What is the Optical TDM in Advanced Multiplexing Strategies? [7]

(b) Briefly explain the Architectures of SONET/SDH. [7]

Q.7

(a) Briefly explain the Network Topologies. Also discuss about the different types of Topologies. [7]

(b) Define the term of Optical detectors. Also state the Physical principles of PIN. [7]

Q.8

(a) Explain the Semiconductor Lasers for Optical Fiber Communication System and their types. [7]

(b) What is the importance of Optical fiber communication in Military application? [7]

Q.9 Write short notes on ***any two*** of the following:- **[7 x 2 = 14]**

(a) Polarization

(b) Single mode fibers

(c) SONET

(d) Avalanche Photodiode

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